# Venkata Lakshmi Parimala Pasupuleti

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AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

## PROFESSIONAL SUMMARY

Staff-level agent orchestration engineer who **built AgentProbe end-to-end** - production LangGraph agent platform with plan-execute-react orchestration, 3-agent runtime (Orchestrator, RAG Tracer, Auto Fix), custom tool calls, multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama), and 5-metric evaluation engine. Designed every layer of the orchestration stack from greenfield to public open-source. 8+ years production Python; 120+ cross-functional sprints at enterprise scale; applied ML in production (1.6M records). M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026). *github.com/venkatalakshmiparimala/AgentProbe*

## CORE COMPETENCIES

Agent Orchestration (LangGraph) | Plan-Execute-React Runtime | Model Orchestration & Routing | Tool Call Architecture | Production Eval & Experimentation | RAG + Vector Search | FastAPI / Python Backend | Multi-Provider LLM Routing | Open-Source Engineering | AWS / Docker / CI/CD | Distributed Backend Systems | Cross-Functional Delivery at Scale

## EXPERIENCE

### Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

**Software Engineer → Associate Software Engineer → Trainee**

Pune, India

- Applied **Python, PyTorch, and OpenAI API** to automate financial data transformation across **1.6M records** - production ML/AI applied to real data workflows at scale.
- Built and delivered **20+ full-stack backend modules** (Python, PostgreSQL, JavaScript) across **120+ Agile sprints** for **100K+ monthly active users**; led cross-functional delivery across Product, Engineering, and QA.
- Led **Docker-based containerization** for 30%+ of modules; improved production query performance by **30%**; maintained **99.5% uptime** for 6+ months via TDD/BDD.
- Built **5+ internal debugging and automation tools** adopted across multiple squads - reducing debugging time by **40%**; designed systems others build on top of.
- Migrated client data to AWS S3 resolving **90% of legacy storage constraints**; encrypted 100% of PHI - zero security incidents.

### George Mason University

Aug 2025 - May 2026

**Graduate Teaching Assistant - Python & Databases**

Fairfax, Virginia

- Mentored 30+ students; designed supplementary exercises; translated complex systems concepts across skill levels - same discipline as enabling teams through platform abstractions.

## PROJECTS

### AgentProbe: Production LangGraph Agent Orchestration Platform OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · Anthropic · OpenAI · Groq · Ollama · Docker · GitHub Actions CI/CD

Staff-level end-to-end ownership of a production agent orchestration platform - every architectural decision from greenfield to public open-source. **Agent runtime**: 3-agent system (Orchestrator, RAG Tracer, Auto Fix) with plan-execute-react loop, state machine management, and custom tool calls. **Model orchestration**: dynamic LLM selection across Anthropic, OpenAI, Groq, Ollama based on latency, capability, and fallback rules. **Execution framework**: handles agent failures, retry logic, and graceful degradation. **Experimentation platform**: 5-metric LLM-as-judge eval engine with per-run scoring, regression tracking, and CI/CD gates. **RAG + vector search**: ChromaDB, embedding pipeline, retrieval quality scoring. **Production backend**: 15-endpoint FastAPI + React observability dashboard. Open-source tool adopted by other engineers. *github.com/venkatalakshmiparimala/AgentProbe*

### AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings

NLP pipeline mapping natural language queries to FHIR APIs via LLMs, BERT, and embeddings with HIPAA-compliant RBAC. **~60% retrieval time reduction.** End-to-end applied AI architecture with domain-specific tool integration. *github.com/venkatalakshmiparimala/AI_on_FHIR*

### CrashClarity: Predictive ML Pipeline - 85%+ Accuracy on 100K+ Records

Python · R · scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages

Full ML lifecycle: feature engineering → model selection → evaluation (MDI + Permutation Importance) → deployment + interactive stakeholder dashboard. **85%+ accuracy.**

## EDUCATION

**M.S. Data Analytics Engineering** - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, ML systems, experimental design, production data pipelines

**B.Tech Computer Science & Engineering** - Sree Venkateshwara College of Engineering 2013-2017

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, TypeScript, SQL, R, Bash, HTML/CSS

**Agent & Orchestration:** LangChain, LangGraph, Plan-Execute-React, Tool Calls, State Machine Management, Multi-Provider LLM Routing (Anthropic, OpenAI, Groq, Ollama), RAG Pipelines, ChromaDB, Prompt Engineering, LLM-as-Judge, LLM Evaluation

**Backend & Infra:** FastAPI, Flask, React, PostgreSQL, MySQL, Docker, AWS (S3, EC2, RDS, Glue), Azure, CI/CD (GitHub Actions), RabbitMQ

**ML & Data:** PyTorch, scikit-learn, BERT, spaCy, Hugging Face, NumPy, Pandas, Feature Engineering, Statistical Modeling

**Process:** Open-Source Development, TDD/BDD, Integration Testing, Git, JIRA, Agile/Scrum